



भारतीय प्रौद्योगिकी संस्थान कानपुर Indian Institute of Technology Kanpur Cognitive Science



Placement Brochure

2026 - 2027

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About Us

What is Cognitive Science ?

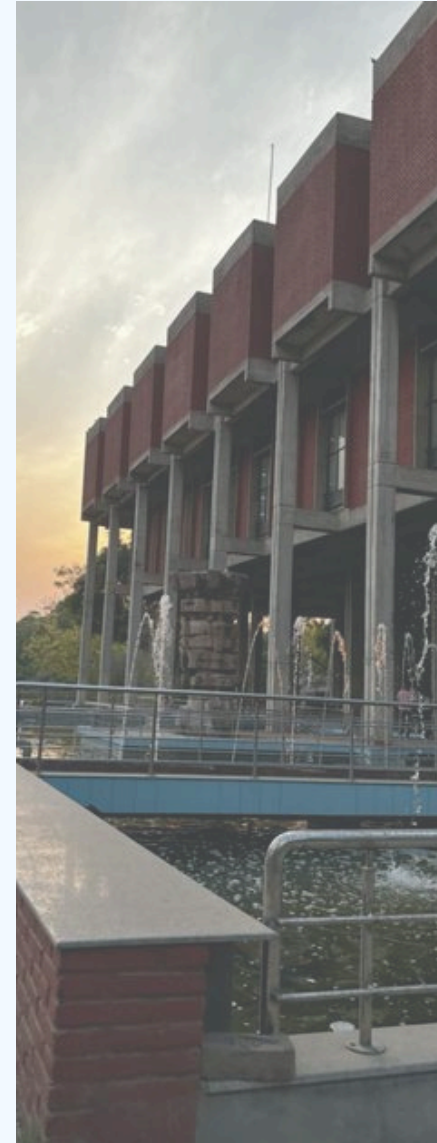
Cognitive Science is the scientific study of intelligence, mind, and machines. At its core, it explores how people perceive, make decisions, think and learn by blending insights from AI, computer science, neuroscience, psychology and linguistics.

What we do ?

Our graduates bridge machine learning and human behavior to build natural, user-aware technology. Trained across mind science, AI, and data analytics, our graduates excel in product design, human-centric UI/UX, explainable AI systems, interpretable ML models, predictive behavioral analytics, natural language models, and adaptive digital interfaces where human insight meets technical performance.

What makes us special ?

Our strength lies in our truly interdisciplinary approach: we combine scientific rigor with practical hands-on application. Rather than focusing on theory alone or building generic algorithms, we apply deep insights into human behavior directly to real-world engineering. This enables us to design intelligent models, intuitive software, and data-driven applications built around the actual needs of real users.



Message from the Head



“

The Cognitive science department is a collective of visionary students who realise that outstanding problems require interdisciplinary thought and application. We believe in a multidimensional outlook: with access to insights from various fields, our students are poised to build more realistic models of human behaviour that are fine-tuned to accommodate real world causal relations.

With use cases in almost every imaginable field, ranging from neuro-marketing, data visualisation, and UX/VR environment design to development of emotionally intelligent and trustworthy AI systems, we host a protean bunch ready to revolutionise our rapidly evolving society.

”

Prof. Devpriya Kumar
PhD

What We Do



NLP & AI

We draw from both linguistics and machine intelligence to build language models, conversational agents, and multilingual AI systems that help machines communicate more like humans.



Data Science and Behavior Analytics

We turn complex behavioral and business data into actionable insights using machine learning, statistical analysis, and predictive modeling. We help organizations understand users, anticipate behavior, and build smarter, data-driven products and strategies.



User Centric Design

We study multisensory perception and neuroscience to design cognitively easy interfaces.

Using cognitive load analysis, eye-tracking, and human-in-the-loop testing, we craft intuitive UI/UX and enable seamless human-technology interaction.



Computational Models of Behavior

We create models that simulate human behavior using Agent-Based Systems, Bayesian Reasoning, and Statistical Decision Models. We predict decision-making, perception, and action, enabling realistic simulations, user modeling, and behavioral forecasting for adaptive systems, recommendation engines, and behavior-aware AI.



Empirical Research

We conduct behavioral and neural research to understand attention, addiction, FOMO, bias, and decision-making.

These insights drive human-centric product design, enabling emotion-aware, habit-forming, and ethically optimized digital experiences.

What Our Students Offer

We combine scientific rigor with practical engineering, making us uniquely suited for roles in AI, HCI, product design, data science and behavioral strategy among many other domains. Here's what we bring to the table in terms of applications.

Data Science & Time-Series Analysis

Develop expertise in data modeling, Bayesian inference, statistical analysis, signal processing, and multimodal data analysis for extracting insights from complex and sequential data.

Language Processing in Humans and Machines

Investigate human and LLM language processing through explainable AI, symbolic approaches, and cognition-inspired NLP to understand and model linguistic knowledge.

Cognitive Ease & Intuitive UI/UX

Design human-centered interfaces using perception science and cognitive load principles to enhance comprehension, reduce user effort, and create intuitive digital experiences.

Cognition-Empowered Software Engineering

Develop cognition-aware software for psychometric testing, cognitive assessment, workload analysis, and human-AI collaboration using full-stack development, behavioral data integration.

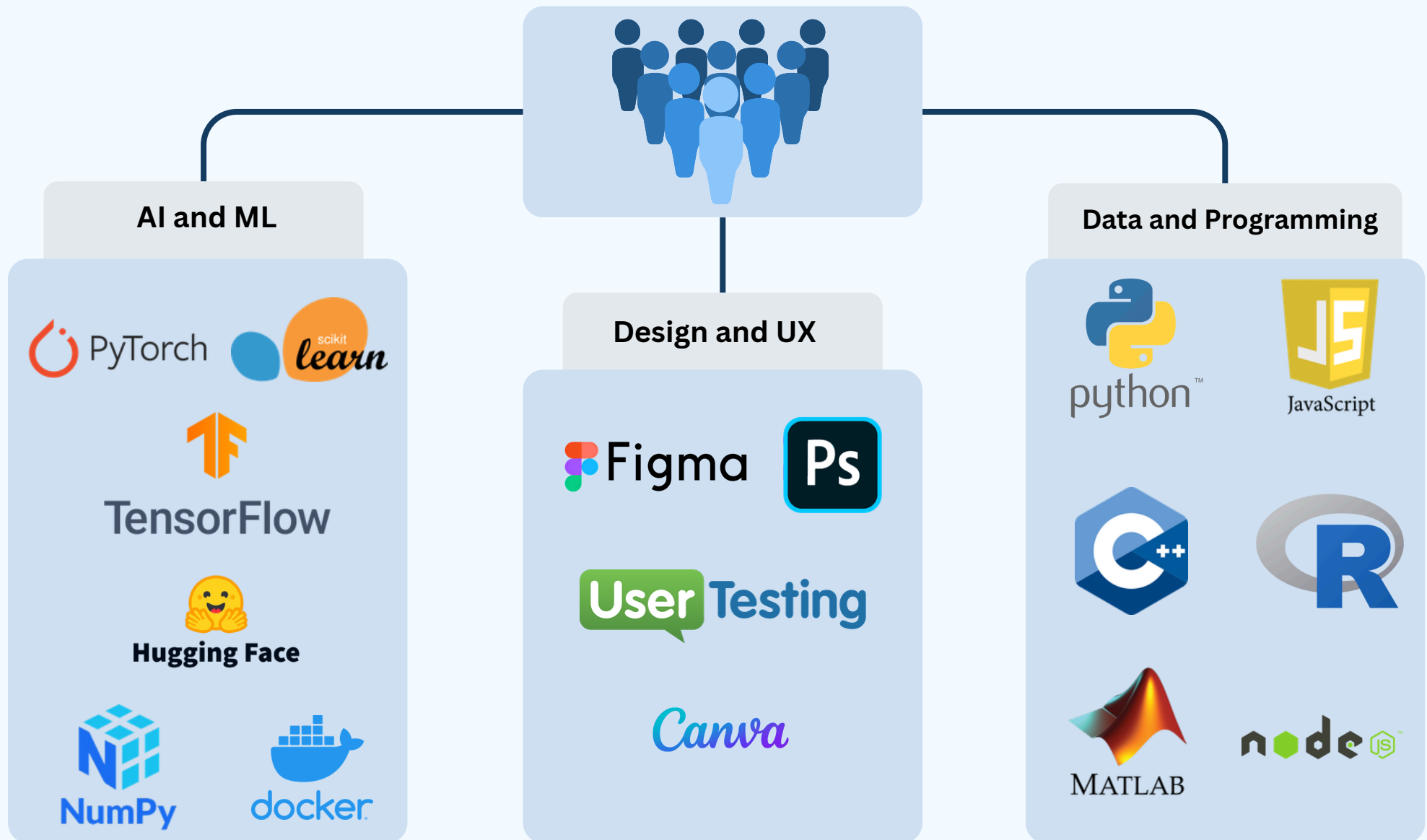
User Behavior Modeling for Businesses

Apply behavioral modeling, decision-making frameworks, and agent-based simulations to understand consumer behavior, optimize business outcomes, drive customer retention and growth.

Strategy & Product Thinking for EdTech

Develop research-driven EdTech solutions that use cognitive learning principles, gamification, and retention-focused content to enhance learning outcomes, motivation, and career readiness.

Skills We Have



Courses

Introduction to ML

Build mathematical and algorithmic foundations for solving complex problems using machine learning.

Big Data and Visual Analytics

Apply scalable algorithms, visualization, and dimensionality reduction to uncover patterns in large datasets

Data Science and Machine Intelligence

Extract insights from data, build predictive models, and apply machine learning to real-world problems

Bayesian Models and Data Analysis

Apply Bayesian inference, build generative models, and analyze real-world data under uncertainty.

Human-Computer Interaction

Design effective technologies by understanding users, tasks, contexts, and interactions

Probabilistic Machine Learning

Model uncertainty and underlying data distributions to make robust probabilistic predictions.

Human-Centered Computing

Design intelligent technologies by integrating computational methods with human needs and behavior

Statistics, Data Analysis and Inference

Apply statistical methods to model and analyze data and conduct hypothesis testing.

Perception Algorithms for Autonomous Systems

Develop algorithms for processing sensory data to enable perception and navigation in autonomous systems

Cognitive Neuroscience

Analyze the neural mechanisms underlying cognition and behavior using computational and analytical approaches.

Computational Cognitive Science

Build computational models of human behavior, decision-making, learning, and cognition.

Deep Reinforcement Learning

Build mathematical and algorithmic foundations for solving complex problems using machine learning.

Experimental Design and Analysis

Design and evaluate empirical experiments using rigorous methodologies and statistical analysis.

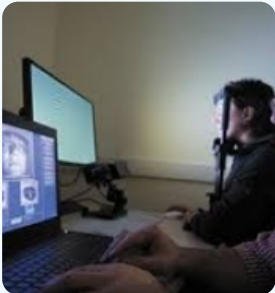
Cognitive Underpinnings of Learning and Technology

Apply cognitive principles of memory, attention, and learning to develop effective educational technologies.

Basics of Psychophysics

Quantify relationships between physical stimuli and human perceptual responses through experimental methods.

Facilities



Language & Eye-Movement Lab

We use high-speed eye-tracking to study reading and gaze behavior in native and non-native speakers.



High Density EEG Lab

We use 128-channel EEG and 8-channel EGG to study brain-body interactions for smarter, human-aware systems.



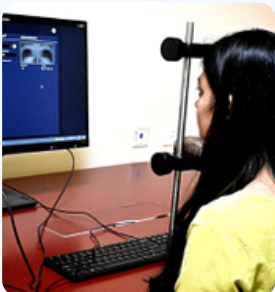
AR/VR Lab

Our AR/VR lab uses immersive technology to study human thought and behavior. We use these tools to study realistic tests on the mind.



Non-invasive brain stimulation lab

We use TMS with neuronavigation to target brain areas using individual MRI data.



Attention & Eye-Movement Lab

We use a 250Hz eye-tracker to study visual attention through pupillometry and psychophysics.



Action lab

We use VR, hand and face tracking to study action, agency, and multimodal integration.

Programs Offered

M.Tech- Master of Technology

2 Years Program

Comprises of rigorous coursework followed by a year of research. Courses often include advanced-level group projects and individual projects.

MSR- Master of Science by Research

2 / 3 Years Program

Similar to the M.Tech Program, with more emphasis on research. Besides fundamental research, students work on many challenging industry-oriented projects.

PhD- Doctor of Philosophy

5 Years+ Program

Highest degree awarded by the department for students interested in research careers. Its focus, unlike other degrees, is more towards generating new knowledge than learning extant knowledge.

L a b s

1] PLATIPUS Lab, Dr. Nisheeth Srivastava

- Computational Cognitive Science
- Probabilistic & Mathematical Modeling
- Data-Driven Behavioral Research
- Interdisciplinary Research

3] TRANSiT Lab, Dr. Pragathi P. Balasubramani

- AI & Machine Learning for Neuroscience
- Real-Time Intelligent Systems
- Brain-Computer Interfaces & Emerging Technologies
- Translational & Industry-Focused Innovation

2] Language Processing Lab - KaLP, Dr. Himanshu Yadav

- Advanced AI & NLP
- Statistical & Probabilistic Modeling
- Human-Centric AI & Cognitive Modeling
- Research & Experimental Skills

4] CLIP Lab, Dr. Anveshna Srivastava

- Cognitive AI & Learning Analytics
- Computational Thinking & Problem Solving
- AR/Technology-Enabled Learning
- Human-Centered Educational Innovation

L a b s

5] CORD Lab, Dr. Sruti S. Ragavan (Adjunct Faculty at Department of Cognitive Science)

- Human–AI Interaction
- Human-Centered AI & HCI
- AI, Software Engineering & Programming
- Technology for India & Social Impact

7] Language & Cognition Lab, Dr. Ark Verma

- Language & Cognition
- Multilingualism & Indian Languages
- Experimental & Computational Methods
- Human-Centered Language Technology

6] Perception, Action & Cognition, Dr. Devpriya Kumar

- Perception & Cognition
- Experimental & Behavioral Research
- Human–Technology Interaction
- Computational & Data-Driven Research

8] Social & Affective Cognition Lab, Dr. K. M. Sharika

- Social & Affective Neuroscience
- Computational & Behavioral Modeling
- Neural & Physiological Analytics
- Human-Centered Research

Faculty

CORE TEAM



Dr. Narayanan Srinivasan
Professor
PHD, UNIVERSITY OF GEORGIA



Dr. Nisheeth Srivastava
Professor
PHD, UNIVERSITY OF MINNESOTA



Dr. Devpriya Kumar
Associate Professor and Head
PHD, Centre of Behavioural and
Cognitive Sciences (CBCS)



Dr. Ark Verma
Associate Professor
PHD, UNIVERSITEIT GHENT



Dr. Himanshu Yadav
Assistant Professor
PHD, UNIVERSITY OF POTSDAM



Dr. Pragathi P. Balasubramani
Assistant Professor
PHD, IIT MADRAS

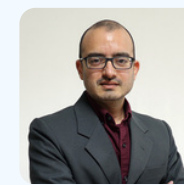


Dr. K. M. Sharika
Assistant Professor
PHD, NATIONAL BRAIN RESEARCH
CENTRE



Dr. Anveshna Srivastava
Assistant Professor
PHD, HBCSE, TIFR, MUMBAI

ADJUNCT FACULTY



Dr. Arjun Ramakrishnan
Professor
PHD, NATIONAL BRAIN RESEARCH
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Dr. Sruti Srinivasa Ragavan
Assistant Professor
PHD, OREGON STATE UNIVERSITY



Dr. Lalit Saraswat
Associate Professor
PHD, IIT BOMBAY



Dr. Nivedita Bhaktha
Assistant Professor
PHD, OHIO STATE UNIVERSITY



Dr. Vivek Kant
Associate Professor
PHD, UNIVERSITY OF WATERLOO

Alumni Footprint



Contact Us



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